

Luciano Ferraro

Senior AI Gameplay Engineer

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Experiences

2022-Current **Senior AI Gameplay Engineer**, *Epic Games*, Fortnite Battle Royale, Lego Fortnite Odyssey, Lego Fortnite Brick Life
C++, Unreal Engine 5

- **Lego Fortnite Odyssey** (Epic🔗 - LEGO🔗) Sole AI engineer for most of the project timeline through prototype and shipping; established the AI architecture. Navmesh, spawning, NPCs, villages, animal taming and friendship system, and many other AI and social systems.
- **Lego Fortnite Brick Life** (Epic🔗 - LEGO🔗) Sole AI engineer through prototype and shipping; owned the AI architecture end-to-end. Navmesh, spawning, job and citizen NPCs, pushed SmartObjects further for this game mode, NPC and Player request system.
- **Sidekicks** (Epic🔗) Authored the prototype of this lightweight, multi-product companion system, later shipped across the Fortnite product line.
- Built the **GenAI/LLM stack for NPCs** in a shipped Fortnite product line: dynamic conversation generation with structured options, moderation, contextual memory, caching, near-zero-delay interactions, runtime TTS voice. Drives in-game minigames, internal monologue, NPC emotions, and SmartObject interactions.
- Daily development with **AI coding agents** across major model families and harnesses (Claude Code, Codex, OpenCode). Authored **Unreal Engine MCP** toolsets exposing UE assets to agents; engineered deferred tool loading (200x context-cost reduction); shipped custom Claude skills widely used internally.
- Ownership of most of the **SmartObject** / **WorldCondition** / **StateTree** stack at Fortnite level. Implemented player usage from scratch and integration with Fortnite AIs; used across the Fortnite ecosystem (Battle Royale, Lego Fortnite, etc.).

2020-2022 **AI Gameplay Engineer**, *Epic Games*, Fortnite Battle Royale
C++, Unreal Engine 4

- **Flying AI (UFO)** (YouTube🔗) First flying AI in Fortnite Battle Royale; custom lightweight 3D navigation system from scratch, full driving/fighting/kidnapping behaviour, custom Octree-based object tracking now reused across Fortnite.
- **Klombo** (YouTube🔗) First big AI players could walk on in Fortnite Battle Royale; lightweight bespoke navmesh with UE4 changes for on-demand geometry, custom collision handling now reused by other AIs and gameplay features.
- Owned **Navmesh in Fortnite**: time-sliced and spatial-based optimizations, dynamic streaming support, shipped a second navmesh at negligible perfmem cost, and daily QA reporting tools that automatically caught most ongoing navmesh issues.
- Built server and client **AI LOD systems**, custom dynamic AI mesh/anim customization with budget-optimizer, took server AI count from tens to 150 concurrent.

2019-2020 **UI Programmer**, *Epic Games*, Fortnite Battle Royale
C++, Unreal Engine 4

- Shipped the ingame matchmaking system, the Feats/Achievements system, and the ingame/frontend map widget. Worked on multiple BattlePasses and the MOTD UI.

2019 **Generalist Programmer**, *Epic Games*, Fortnite Battle Royale
C++, Unreal Engine 4

- Replay system work including a custom "Hero's Path"-style camera mode. Contributed to the split-screen initiative.
- 2017-2019 **AI Game Programmer**, *Build a Rocket Boy*, Everywhere Game
C++, Lua, Lumberyard/Open 3D Engine
buildarocketboy.com - everywhere.game
- Main AI Engineer on a 3rd person shooter, openworld multiplayer game. Built AI driving and walking on a Road Network (roads, sidewalks, traffic lights, crossings) with Navigation, Spawning and Behaviour systems on top.
 - AI Reaction System; SmartObjects usage by AIs; cover generation system (also used by players); multiple combat AI types with complex tactical behaviours.
- 2017-2017 **AI Game Programmer**, *LVGameDev LLC*, Consulting, C#, Unity
Integrated ReGoap in their Unity game: SimAirport.
- 2013-2018 **Full-stack developer**, *Ferraro Technology SRLS*
Owner of a small web development company.
- 2010-2013 **Full-stack developer**, Freelance
Multiple small websites as a contractor.

Skills

GenAI / LLM	Conversation generation, structured options, moderation, contextual memory , near-zero-delay LLM-driven interactions, TTS, emotion/feeling deltas. Local Llama and direct LLM API experimentation.
Agentic Development	Claude Code, Codex, OpenCode harnesses; MCP server plugins, custom Claude skills . Claude (Opus/Sonnet/Haiku), GPT/Codex and other model families. Deferred tool loading.
AI Behaviours	Behavior Trees, StateTrees, GOAP, SmartObjects, WorldConditions , FSM, behavior injection. Author of ReGoap.
AI Navigation	Navmesh, custom 3D navigation, navlinks , custom graph navigation, dynamic invokers, road-network navigation.
AI Performance	Server and client AI LOD, dynamic AI relevancy , perfmem reporting, evaluator and tick budgeting.
Gameplay	Animation, Conversation, Persistence, Replication / Push Model , Physics integration.
Game Engines	Unreal Engine 5 , Unity3D, Godot, Lumberyard / Open 3D Engine.
Languages	C++/C, C#, Python ; Java, JavaScript.
Consoles	PS4, PS5, Xbox One/Series, Switch .
Networking	UE5, Push Model Networking , custom protocols.
Backend / Web	PostgreSQL, MySQL, MongoDB, Redis ; Django, Flask; Nginx, Apache.
Spoken Language	Italian (native), English (full professional proficiency) .

F/OSS

Author **ReGoap**

GitHub: github.com/luxkun/ReGoap

Engine-agnostic C# GOAP library with Unity and Godot adapters. 1.1k+ stars and 150+ forks on GitHub.

Education

Bachelor's degree in Computer Science, University of "La Sapienza"

Certificate of Attendance, Game developing school "AIV"